

derivative

$$f(x) = 3x^2$$

$$\frac{d}{dx} f = 6x$$



```
function partial (func, varName, point) {
```

```
  const dx = 0.00001
```

```
  const { [varName]: x0 } = point
```

```
  const df = func ( { ...point, [varName]: x0 + dx } ) -  $\frac{\text{func (point)}}{dx}$ 
```

```
  return dx
```

```
}
```

$$\frac{f(x + dx) - f(x)}{dx}$$

dx

I find myself Discovering what appears as alien code

I know what it does BUT NOT how

$$\nabla f(x) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$\nabla \cdot$ Divergence

$\nabla \times$ curl